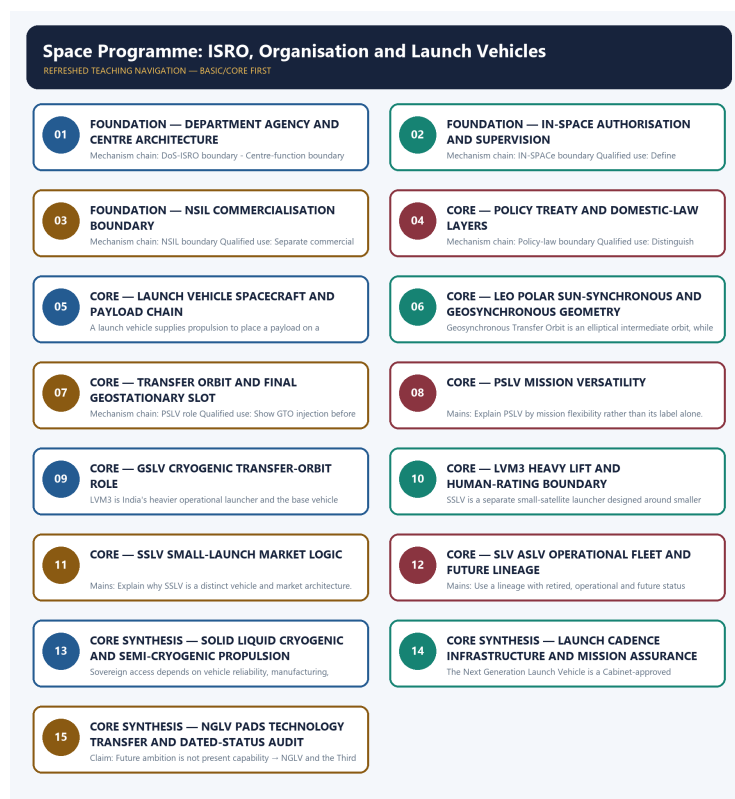


SOLVED PRACTICE WORKBOOK

Space Programme: ISRO, Organisation and Launch Vehicles - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Space Programme: ISRO, Organisation and Launch Vehicles - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies DoS-ISRO boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: A. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of DoS-ISRO boundary?

A. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: B. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses DoS-ISRO boundary without changing its institution, unit or status?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: C. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about DoS-ISRO boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: D. Explanation: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Centre-function boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: A. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Centre-function boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: B. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Centre-function boundary without changing its institution, unit or status?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: C. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not

transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Centre-function boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.

Answer: D. Explanation: VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies IN-SPACE boundary?

A. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: A. Explanation: IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of IN-SPACE boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: B. Explanation: IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses IN-SPACE boundary without changing its institution, unit or status?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. IN-SPACE is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: C. Explanation: IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about IN-SPACe boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: D. Explanation: IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies NSIL boundary?

A. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: A. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of NSIL boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: B. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses NSIL boundary without changing its institution, unit or status?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.

Answer: C. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about NSIL boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: D. Explanation: NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Policy-law boundary?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: A. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Policy-law boundary?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: B. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Policy-law boundary without changing its institution, unit or status?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.

Answer: C. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Policy-law boundary?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Answer: D. Explanation: Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Launcher-spacecraft boundary?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: A. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Launcher-spacecraft boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: B. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Launcher-spacecraft boundary without changing its institution, unit or status?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: C. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Launcher-spacecraft boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.

Answer: D. Explanation: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Orbit-family boundary?

A. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: A. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Orbit-family boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: B. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms

for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Orbit-family boundary without changing its institution, unit or status?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: C. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Orbit-family boundary?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.

Answer: D. Explanation: LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies GTO-GEO boundary?

A. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: A. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of GTO-GEO boundary?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: B. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the

spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses GTO-GEO boundary without changing its institution, unit or status?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: C. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about GTO-GEO boundary?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.

Answer: D. Explanation: Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies PSLV role?

A. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: A. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of PSLV role?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: B. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses PSLV role without changing its institution, unit or status?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: C. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about PSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.

Answer: D. Explanation: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies GSLV role?

A. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of GSLV role?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: B. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses GSLV role without changing its institution, unit or status?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: C. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about GSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.

Answer: D. Explanation: GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies LVM3 role?

A. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of LVM3 role?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: B. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses LVM3 role without changing its institution, unit or status?

A. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: C. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about LVM3 role?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.

Answer: D. Explanation: LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies SSLV role?

A. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: A. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of SSLV role?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: B. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses SSLV role without changing its institution, unit or status?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: C. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about SSLV role?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Answer: D. Explanation: SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Lineage-status boundary?

A. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: A. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Lineage-status boundary?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: B. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Lineage-status boundary without changing its institution, unit or status?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: C. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Lineage-status boundary?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. D. SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.

Answer: D. Explanation: SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Propulsion-class boundary?

A. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: A. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Propulsion-class boundary?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: B. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Propulsion-class boundary without changing its institution, unit or status?

A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: C. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Propulsion-class boundary?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Answer: D. Explanation: Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Human-rating boundary?

A. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: A. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Human-rating boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Human-rating boundary without changing its institution, unit or status?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: C. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Human-rating boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

Answer: D. Explanation: Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Launch-system boundary?

A. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: A. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Launch-system boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Launch-system boundary without changing its institution, unit or status?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: C. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Launch-system boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.

Answer: D. Explanation: Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies NGLV status boundary?

- A. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: A. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of NGLV status boundary?

- A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: B. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses NGLV status boundary without changing its institution, unit or status?

- A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: C. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about NGLV status boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.

Answer: D. Explanation: The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Infrastructure-status boundary?

A. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.

Answer: A. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Infrastructure-status boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: B. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Infrastructure-status boundary without changing its institution, unit or status?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: C. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Infrastructure-status boundary?

A. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.

Answer: D. Explanation: The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Technology-transfer boundary?

A. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: A. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Technology-transfer boundary?

A. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. B. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. C. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. D. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.

Answer: B. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Technology-transfer boundary without changing its institution, unit or status?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. C. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: C. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Technology-transfer boundary?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. C. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. D. Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Answer: D. Explanation: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile launch-data boundary?

A. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. B. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

Answer: A. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile launch-data boundary?

A. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. B. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. C. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: B. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile launch-data boundary without changing its institution, unit or status?

A. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. B. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. C. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. D. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.

Answer: C. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile launch-data boundary?

A. A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. B. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. C. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. D. Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Answer: D. Explanation: Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2018 PSLV-GSLV objective concept and a provisional 2026 IN-SPACe/private-participation concept here. No unverified objective key or direct Mains demand is invented.

PYQ DEMAND CARD 1 - 2018 Prelims GS-I

Demand: Compare PSLV and GSLV as Indian satellite launch vehicles.

Status: Verified routed objective concept; the official key was unavailable locally, so no answer letter is asserted.

Model solution: Launcher-spacecraft boundary: A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **GTO-GEO boundary:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **PSLV role:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **GSLV role:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2026 Prelims GS-I

Demand: Private-sector participation, IN-SPACe mandate and Indian space-startup achievements.

Status: Verified provisional routed concept; the local key is provisional and no answer letter is recorded or inferred.

Model solution: DoS-ISRO boundary: The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **IN-SPACe boundary:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **NSIL boundary:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Policy-law boundary:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Technology-transfer boundary:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish the roles of DoS, ISRO, IN-SPACe and NSIL. Answer in about 150 words.

Model thesis: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.
- VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it.
- IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.

- NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.
- Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Qualified conclusion: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Centre-function boundary. **Named evidence/example:** VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate PSLV, GSLV, LVM3 and SSLV without reducing the comparison to payload size. Answer in about 150 words.

Model thesis: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for

Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.
- Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.
- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

Qualified conclusion: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PSLV role.

Named evidence/example: PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain how propulsion and orbit requirements shape launcher choice. Answer in about 250 words.

Model thesis: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation.
- LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit.
- Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot.
- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.

Qualified conclusion: Claim: Launcher-spacecraft boundary. **Named evidence/example:** A launch vehicle supplies propulsion to place a payload on a trajectory or into an orbit, whereas the spacecraft or satellite carries the mission payload and may use its own propulsion after separation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Orbit-family boundary. **Named evidence/example:** LEO, polar orbit, sun-synchronous orbit, geosynchronous orbit and geostationary orbit describe different geometry, altitude or timing properties; polar and sun-synchronous are not synonyms for every low orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GTO-GEO boundary. **Named evidence/example:** Geosynchronous Transfer Orbit is an elliptical intermediate orbit, while geostationary orbit is circular and equatorial; a launcher injection into GTO does not by itself place the spacecraft in its final geostationary slot. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess Indian space reform as institutional differentiation rather than privatisation. Answer in about 250 words.

Model thesis: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law

boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable.
- IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm.
- NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision.
- Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Qualified conclusion: Claim: DoS-ISRO boundary. **Named evidence/example:** The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IN-SPACe boundary. **Named evidence/example:** IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NSIL boundary. **Named evidence/example:** NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Policy-law boundary. **Named evidence/example:** Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's access-to-space ecosystem through vehicles, infrastructure, industry and mission assurance. Answer in about 300 words.

Model thesis: **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit.
- GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together.
- LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses.
- SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant.

- Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.
- The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.
- Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Qualified conclusion: **Claim:** PSLV role. **Named evidence/example:** PSLV is the versatile operational workhorse used across Earth-observation, navigation and multi-payload missions; its name does not confine every mission to a polar orbit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GSLV role. **Named evidence/example:** GSLV uses an indigenous cryogenic upper stage for heavier communication-class missions to transfer orbit; cryogenic capability, orbit and payload class must be stated together. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** LVM3 role. **Named evidence/example:** LVM3 is India's heavier operational launcher and the base vehicle selected for human rating for Gaganyaan; an operational launcher and its human-rated configuration are distinct statuses. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SSLV role. **Named evidence/example:** SSLV is a separate small-satellite launcher designed around smaller payloads and quicker, lower-infrastructure launch demand; it is not a PSLV variant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a status-disciplined roadmap from the current fleet to future launch capacity. Answer in about 300 words.

Model thesis: Claim: Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Human-rating boundary. **Named evidence/example:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NGLV status boundary. **Named evidence/example:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary. **Named evidence/example:** Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column.
- Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage.
- Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record.

- Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence.
- The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance.
- The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure.
- Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.
- Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date.

Qualified conclusion: Claim: Lineage-status boundary. **Named evidence/example:** SLV-3 and ASLV are historical technology-learning vehicles, PSLV, GSLV and LVM3 are identified in the repository owner as operational, and approved future systems must not be placed in the same status column. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Propulsion-class boundary. **Named evidence/example:** Solid, earth-storable liquid, cryogenic liquid-hydrogen and semi-cryogenic kerosene systems differ in propellants and operating role; a semi-cryogenic development programme is not a flying cryogenic stage. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Human-rating boundary. **Named evidence/example:** Human rating adds margins, redundancy, health monitoring, benign failure behaviour and abort capability to a launcher system; it is a qualification process, not merely a strong flight record. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-system boundary. **Named evidence/example:** Sovereign access depends on vehicle reliability, manufacturing, integration bays, pads, range safety, propellant logistics and mission assurance; a rocket design alone does not establish launch cadence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NGLV status boundary. **Named evidence/example:** The Next Generation Launch Vehicle is a Cabinet-approved development programme with a planned reusable or recoverable first-stage architecture; approval is not a flight, operational service or demonstrated payload performance. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:**

Infrastructure-status boundary. **Named evidence/example:** The Third Launch Pad and Kulasekarapattinam launch complex belong to future capacity expansion; approved or under-construction infrastructure is not commissioned infrastructure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Technology-transfer boundary.

Named evidence/example: Transfer of SSLV technology to HAL is an industrial production and commercialisation arrangement; it does not mean that ISRO, IN-SPACe, NSIL and the State's safety responsibilities disappear.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile launch-data boundary. **Named evidence/example:** Payload capacity, configuration, launch outcome, fleet status, project cost, schedule and anomaly findings require a dated official ISRO, Department of Space or Cabinet source and must never be carried forward without that date. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.